Prithvi Raj

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Education

Indian Institute of Information Technology, Guwahati Aug 2023 – Present
B.Tech in Electronics and Communication Engineering **CPI:** 7.73 **SPI:** 8.71

Research & Core Projects

Distributed Event-Driven Microservices Platform [Github] June 2026 - Present

- Architected a distributed event-driven microservices platform using Kafka. Also Containerized multi-service infrastructure, configuring isolated consumer groups and dead-letter queues (DLQ) for idempotent processing.
- Implemented Spring Cloud Gateway with Redis-backed rate limiting, Resilience4j circuit breakers, and Redis multi-layer caching to optimize DB read throughput under peak load.

Project Ascend: AI-Powered Life Tracker [Github] July 2025

- Programmed a full-stack wellness tracker using React, gamifying daily habits with AI-generated side-quests.
- Logged activities and generated weekly reports via a modular backend with JWT auth and PostgreSQL.

Multi Level RAG Pipeline for Document Querying [Github] July 2025

- Orchestrated an advanced Retrieval-Augmented Generation (RAG) pipeline for domain-specific QA using multi-LLM orchestration, Chain-of-Thought (CoT) prompting, and agentic workflows.
- Integrated adaptive chunking strategies, custom re-ranking algorithms, and a feedback-driven evaluation module that autonomously re-runs queries until reaching over 65% accuracy, reducing hallucination rates by 30%.

RoundUpX: Microservices Financial Savings Platform [Github] March 2025 - April 2025

- Architected scalable RESTful microservices in Go to process real-time UPI transaction round-ups, defining clean API contracts, schemas, and background job workers for asynchronous event handling.
- Integrated third-party payment APIs, implementing idempotency keys, exponential backoff retries, and transactional reconciliation logic to reliably manage webhook callbacks and edge-case status codes.
- Engineered automated CI/CD pipelines via GitHub Actions to run unit tests and deploy containerized Docker services to AWS cloud environments with minimal downtime.
- Instrumented structured logging and distributed tracing patterns across endpoints to streamline production debugging, partner response verification, and metric tracking.

Experience & Leadership

Semiconductor Workforce Development Program *TCFST & NTHU, Taiwan* March 2026

- Executed 2-micrometer precision multi-layer silicon wafer fabrication via hands-on cleanroom operations at National Tsing Hua University, achieving over 90% yield for critical layers.
- Designed and tested four new Photolithography processes, leading to a significant reduction in production waste by 30%; techniques are now adopted as standard practices across the fabrication teams in the collaboration.

Research Intern (SPARC Program) *IIT Hyderabad* Nov 2025

- Fabricated a functional Resistive Gas Sensor detecting different gas levels, specifically optimizing for Ammonia.
- Successfully executed multi-layer semiconductor fabrication with 10-micrometer detail.

Coordinator *Alankrit Fine-Arts Club – IIIT Guwahati* Dec 2024 – July 2025

- Led 5+ campus-wide arts events, managing logistics; recognized as the club's most active coordinator.

Technical Skills

Languages: Go (Golang), C++, Python, JavaScript, SQL, JAVA
Systems & Backend: REST APIs, Idempotency & Retries, Background Jobs, Multi-threading, Win32 API
Cloud & Infrastructure: AWS, GCP, Kubernetes, Docker, Apache Kafka, Redis, CI/CD Pipelines, PostgreSQL
AI & Frameworks: SpringBoot, PyTorch, 3D Gaussian Splatting(3DGS), OpenCV, Scikit-learn, Pandas, NumPy

Achievements

Hackathon Finalist, India International Science Festival (IISF) 2024

- Recognized as a finalist team in the top 0.2% (from 45,000+ competing teams) in the national-level S&T Hackathon.

First Prize, GitHub University Blind Coding Challenge 2023

- Ranked #1 among all first-year students across every engineering discipline in a university-wide competition.